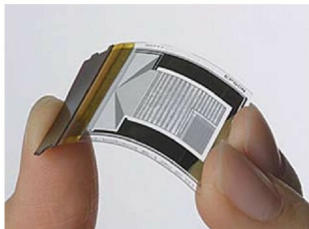


Asynchronous Microprocessor

The Clockless Future

The synchronous processor chips of computers use a clock to time the entire chip. But this clock is also the cause of speed and power consumption headaches. Here is how an asynchronous chip overcomes these problems by eliminating the clock altogether



World's first flexible 8-bit asynchronous microprocessor developed by Seiko Epson using low-temperature polysilicon thin-film transistors on a plastic substrate. With energy consumption reduced by 70 per cent compared to the synchronous microprocessors now in everyday use, Epson is now researching potential applications for its invention (courtesy: <http://global.epson.com>)

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The growing demand for high-performance, time-critical battery-powered devices like cellphones and personal digital assistants has made the manufacturers revisit the old concept of clockless, asynchronous microprocessor design. Asynchronous processors, also called clockless or self-timed, don't use the oscillating crystal that serves as the regularly 'ticking' clock to speed up the work done by traditional synchronous processors.

While the central clock in synchronous processors causes prob-

lems of clock skew, power efficiency and electromagnetic compatibility, asynchronous processors work without waiting for a clock tick, with very low electronic interference. The resulting reliability and robustness of asynchronous microprocessor chips have made them popular research subjects for applications such as pagers, smart cards and cellphones. In this article, we detail the working of asynchronous processors.

Synchronous vs asynchronous processors

Today, synchronous processors consist of hundreds of millions of transistors and reach clock rates up to several gigahertz (GHz). The high clock rates imply that several million transistors switch several billion times a second, regardless of work to be done or not, simply because of the clock signal. Up to 40 per cent of the power consumption is spent in clock generation, the drivers and the clock tree.

Another problem of synchronous processors is the clock skew, which is of growing concern with increasing die sizes and higher clock rates. For

distributing the clock signal on the chip in a manner such that it reaches all sinks at exactly the same point in time, a huge effort must be made. For example, the H-clock tree or B-Tree can achieve a low clock skew at the cost of many additional wires, which increases the chip size, cost and energy dissipation due to the capacitances that are created by thousands of parallel wires in the clock tree.

The speeding up of a synchronous processor is possible by optimising the critical path. Either its elements have to be optimised or it has to be split up so that it takes more than one clock cycle to compute the final result. The latter way introduces a new overhead.

But, in asynchronous processors, there are several starting points: Each optimised element can speed up the processor, so there is no need to make effort to optimise only rarely used parts. Instead, only frequently used parts are optimised to increase the average speed of the entire processor.

If changes in combinatory logic elements (like increase in temperature and lowered supply voltage) slow down the operational performance of the processor, the clock of the synchronous chip has to be regulated actively to keep the system in a valid state. But, asynchronous systems can implicitly adapt to these variations, which means that they always run at the maximum speed possible. Another implication of the automatic performance adaptation is that an asynchronous system always